

AWS DEPLOYMENT GUIDE

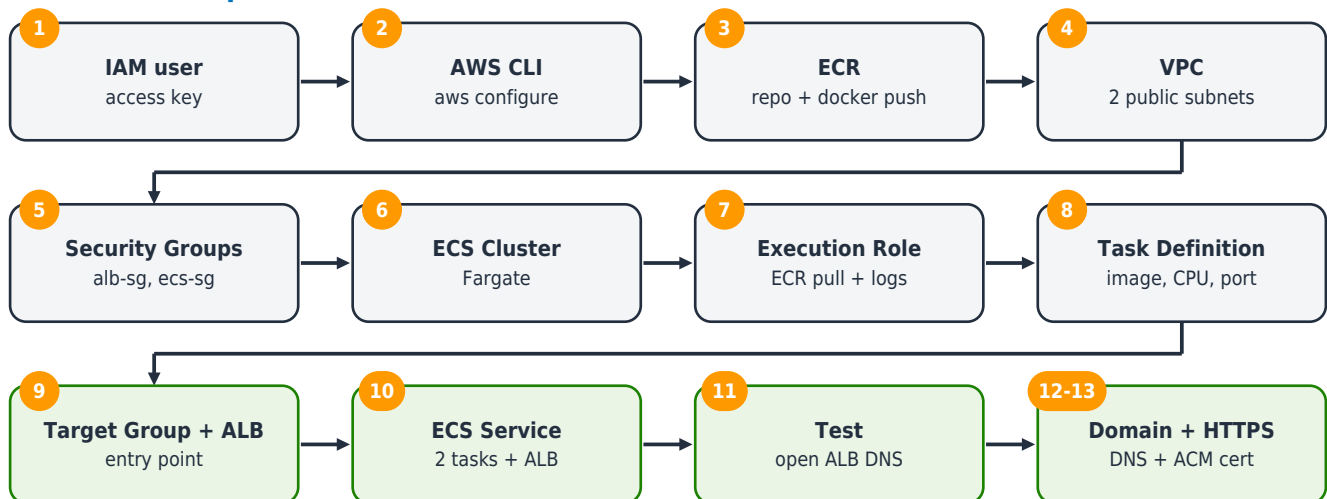
Express App ko AWS par Live karo

Docker image → ECR → ECS Fargate → Load Balancer → Apna Domain + HTTPS

Zero se live tak: har click, har command, har "kyun?" ka jawab

Kya banayenge	Express (Node.js) app, Docker container me, 2 copies AWS Fargate par, ALB ke peeche, apne domain par
Region	Asia Pacific (Mumbai) → code ap-south-1 . Saari cheezein isi region me banani hain
Time	Pehli baar lagbhag 60-90 minute (ALB / ECS ka wait time milake)
Cost warning	ALB, Fargate tasks aur public IPv4 hourly bill banate hain. Kaam khatam hone par Cleanup section zaroor follow karo

Poora Roadmap (ek nazar me)



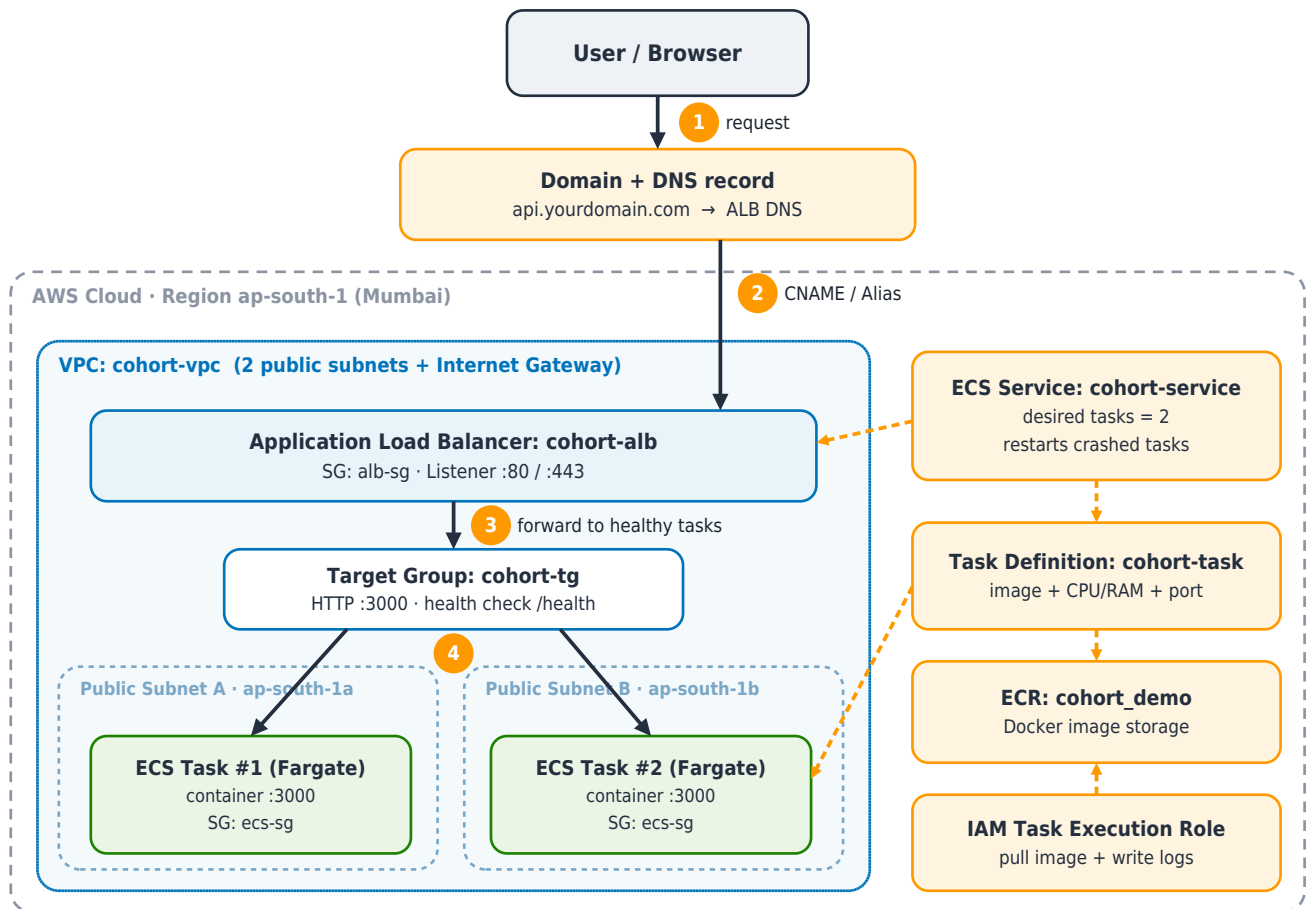
Neeche har step number upar ke roadmap se match karta hai. Orange-left-border wale boxes **"Notes me sudhaar"** hain (jahan tumhare notes me galti / kami thi), aur neele boxes **"Kyun?"** hain (tumhare sawaalon ke jawab).

Pehle Concepts Samjho

Commands se pehle ek baar ye samajh lo ki har AWS service kya kaam karti hai. Phir steps khud samajh aayenge.

————→ solid = user traffic

- - - - -> dashed = ECS control (manage / launch)



Request ka safar (diagram ke numbers)

- 1 User browser me **api.yourdomain.com** kholta hai.
- 2 DNS record us naam ko **ALB ke DNS name** se jod deta hai (CNAME ya Alias).
- 3 ALB request leta hai (port 80/443) aur **Target Group** ke zariye kisi **healthy task** ko bhejta hai.
- 4 Task ke andar tumhara Express container port **3000** par response banata hai, jo wapas usi raaste se user tak jata hai.

Peeche-peeche (dashed orange): **ECS Service** dekhti rehti hai ki hamesha 2 tasks chal rahe hon; **Task Definition** batati hai kaunsa image (ECR se), kitni CPU/RAM, kaunsa port; **Execution Role** ECS ko permission deta hai image pull karne aur logs likhne ki.

Kaun kya karta hai (cheat table)

Service	Kaam	Real-life example
IAM	Permission system: kaun kya kar sakta hai	Building ke ID card + security guard
ECR	Docker images ka private godown (storage) . Sirf store karta hai, run nahi	Docker Hub ka private AWS version
ECS	Container manager : decide karta hai kaunsa container kahan chalega, crash hone par restart	Restaurant manager
Fargate	ECS ka serverless mode: server (EC2) khud manage nahi karna padta	Cloud kitchen: kitchen AWS ka, recipe tumhari
Cluster	Tasks ka logical group (khud me free)	Folder
Task Definition	Blueprint: image, CPU, RAM, port, env vars, role	Recipe card
Task	Task Definition ki ek chalti hui copy (running container)	Recipe se bani ek dish
Service	Guarantee: desired count (2) tasks hamesha chalein; ALB se jodta hai; naya version rolling deploy karta hai	Shift supervisor
VPC / Subnet	AWS ke andar tumhara private network / uske hisse (har AZ me ek)	Society / uske blocks
Security Group	Firewall: kaunsa traffic kis port par andar aa sakta hai	Gate par guard ki list
ALB	Internet se aane wali traffic leta hai aur healthy tasks me baant deta hai	Reception desk
Target Group	ALB ke peeche ke targets (tasks) ki list + health check	Staff list: present / absent
Route 53 / DNS	Domain naam → ALB ka address	Phone book
ACM	Free SSL certificate (HTTPS ka padlock)	Website ka ID proof
CloudWatch Logs	Container ke logs yahin dikhte hain	App ki diary

NOTES ME SUDHAAR

Notes me likha tha "ecs = jo actual server run karta hai" aur "service = final image run karti hai". Thoda sa sudhaar: ECS khud **manager** hai, container asal me **Fargate** (ya EC2) par chalta hai. Aur Service image nahi chalati: wo **Task Definition ke hisaab se tasks chalati aur zinda rakhti hai**.

Prerequisites (shuru karne se pehle)

Cheez	Detail
AWS account	Card verified. Root user par MFA ON kar do (Security credentials page)
Docker Desktop	Install karke running rakho. Check: <code>docker --version</code>
Terminal	Warp / PowerShell / Git Bash / Mac Terminal: koi bhi chalega
Project	Dockerfile ready, app port 3000 par, ek /health route (ALB health check ke liye)
Domain	Step 12 tak chahiye (abhi nahi liya to koi baat nahi, ALB DNS se test ho jayega)

Express app me ye hona chahiye (health route + port 3000):

```
app.get('/health', (req, res) => res.status(200).send('OK'));
app.listen(3000, () => console.log('Server running on 3000'));
```

Dockerfile (project root me):

```
FROM node:20-alpine
WORKDIR /app
COPY package*.json ./
RUN npm ci --omit=dev
COPY . .
EXPOSE 3000
CMD ["node", "index.js"]
```

Aur ek **.dockerignore** file bana do: `node_modules`, `.env`, `.git`.

⚠ DHYAN DO

.env ko image me kabhi mat daalo. Secrets (DB URL, API keys) Task Definition ke environment variables ya AWS Secrets Manager se do (Step 8). Agar app `localhost` / `127.0.0.1` par explicitly listen karti hai to ALB use nahi pahunch payega. Default `app.listen(3000)` theek hai.

1 IAM User banao (root use mat karo)

KYUN? (Why)

Root user ke paas poore account + billing ka unlimited access hota hai. Uski key leak hui to account aur paisa dono gaye, aur use limited bhi nahi kar sakte. **IAM user** ko sirf zaroori permissions milti hain aur key leak ho to turant delete kar sakte ho.

- 1 AWS Console me upar search box me **IAM** likho → open karo.
- 2 Left menu → **Users** → **Create user**.
- 3 User name: `cohort_user` → **Next**. (Console login chahiye to "Provide user access to the AWS Management Console" tick karo; sirf CLI ke liye zaroori nahi.)
- 4 Permissions options: **Attach policies directly**.
- 5 Neeche table ki policies search karke tick karo → **Next** → **Create user**.
- 6 User kholo → **Security credentials** tab → **Create access key** → use case **Command Line Interface (CLI)** → confirmation tick → **Next** → **Create access key**.
- 7 **Access key ID** aur **Secret access key** copy karke safe jagah rakho (ya **Download .csv**). Secret sirf **ek hi baar** dikhta hai.

Policy	Kyun chahiye	Zaroori?
AmazonEC2ContainerRegistryFullAccess	ECR repo banana, docker push / pull	Haan
AmazonECS_FullAccess	Cluster, task definition, service	Haan
AmazonEC2FullAccess	VPC, subnets, security groups, ALB, target group (ye sab EC2 service ke andar aate hain)	Haan
IAMFullAccess	Step 7 ka Execution Role banana. (Ya wo role root / admin se bana lo)	Agar role <code>cohort_user</code> se banana ho
AmazonRoute53FullAccess	Route 53 me DNS record	Domain Route 53 par ho to
AWSCertificateManagerFullAccess	HTTPS certificate (Step 13)	Step 13 ke liye
CloudWatchLogsReadOnlyAccess	Container logs dekhna	Optional

NOTES ME SUDHAAR

Notes me sirf "ec2, ecs full access" likha tha. **ECR ki policy missing thi**: uske bina `docker push` par AccessDenied aata. Full-access policies sirf learning ke liye hain; production me least privilege use karte hain.

⚠ DHYAN DO

Access key **kabhi Git me commit / WhatsApp / screenshot me share mat karo**. Leak ho jaye to IAM → user → Security credentials → key **Delete**.

2 AWS CLI install karo + aws configure

- 1 AWS CLI v2** install karo (Windows: MSI installer, Mac: pkg installer, aws.amazon.com/cli se). Check: `aws --version`
- Terminal kholo (Warp / PowerShell / Git Bash / Terminal) aur `aws configure` chalao.
- Prompts me values daalo (neech).
- Verify karo ki sahi user se login hua: `aws sts get-caller-identity`

```
$ aws configure
AWS Access Key ID [None]: AKIA.....
AWS Secret Access Key [None]: *****
Default region name [None]: ap-south-1
Default output format [None]: json

$ aws sts get-caller-identity
{
  "UserId": "AIDA.....",
  "Account": "123456789012",      <-- ye 12 digit ACCOUNT ID aage kaam aayegi
  "Arn": "arn:aws:iam::123456789012:user/cohort_user"
}
```

KYUN? (Why)

Warp terminal kyun? Zaroori **nahi** hai. Warp sirf ek modern terminal hai (commands ke blocks, easy copy, AI suggestions). Windows par PowerShell / CMD / Git Bash ya Mac par Terminal bhi bilkul chalega: commands same hain. Bas Docker aur AWS CLI terminal me chalne chahiye.

✓ TIP

Console ke **top-right region dropdown** me hamesha **Asia Pacific (Mumbai)** select rakho. Agar ECR Mumbai me aur ECS Oregon me bana diya to kuch dikhega hi nahi.

3 ECR repository banao + Docker image push karo

- Console → search **ECR** (Elastic Container Registry) → **Repositories** → **Create repository**.
- Visibility: **Private**. Repository name: `cohort_demo`. Baaki defaults rehne do → **Create**.
- Repo kholo → upar **View push commands**: yahan 4 commands dikhte hain (Mac/Linux tab aur Windows tab).
- Terminal me **project folder me jao** (jahan Dockerfile hai): `cd path/to/your/project`
- Neeche wale 4 commands chalao. `<ACCOUNT_ID>` ki jagah apna 12 digit account id daalo.

```
# 1) ECR login (Docker ko ECR ka temporary token milta hai)
aws ecr get-login-password --region ap-south-1 | docker login --username AWS \
  --password-stdin <ACCOUNT_ID>.dkr.ecr.ap-south-1.amazonaws.com

# 2) BUILD: console wala "docker build -t cohort_demo ." SKIP karo, ye chalao:
docker buildx build --platform linux/amd64 -t cohort_demo:latest . --load

# 3) TAG (local image ko ECR ka naam do)
docker tag cohort_demo:latest <ACCOUNT_ID>.dkr.ecr.ap-south-1.amazonaws.com/cohort_demo:latest

# 4) PUSH
docker push <ACCOUNT_ID>.dkr.ecr.ap-south-1.amazonaws.com/cohort_demo:latest
```

KYUN? (Why)

2nd command kyun skip? Normal `docker build` tumhare laptop ki CPU ke hisaab se image banata hai. Apple Silicon Mac (M1/M2/M3) par wo **arm64** hoti hai, jabki hum Fargate par **x86_64 (amd64)** chala rahe hain. Mismatch ho to task start hote hi crash: `"exec format error"`. `--platform linux/amd64` x86 image force karta hai, aur `--load` us image ko local Docker me rakhta hai taaki `docker tag` kaam kare.

NOTES ME SUDHAAR

Notes me `"--patform"` likha tha. Do galtiyan: spelling **platform** hai, aur dash **do normal hyphens (--)** hain. Phone / Word me `--` auto-correct hokar lambi dash (`—`) ban jaata hai aur command fail ho jaata hai. Command seedha terminal se type karo.

KYUN? (Why)

Windows par Git Bash kyun? Console ke **Windows tab** ke commands PowerShell-specific hain (AWS Tools for PowerShell chahiye). Git Bash ek bash shell hai, isliye wahan **Mac / Linux tab ke commands** chalte hain. Ya to Windows tab + PowerShell, ya Mac/Linux tab + Git Bash: dono me se ek hi rakho.

✓ TIP

Push ke baad ECR → cohort_demo me image dikhni chahiye, tag **latest** ke saath. ECR login token ~12 ghante chalta hai; "no basic auth credentials" aaye to command 1 dobara chalao.

4 VPC banao (apna private network)

- 1 Console → **VPC** → **Create VPC**.
- 2 Resources to create: **VPC and more**.
- 3 Name tag auto-generation: **cohort** (naam bante hain cohort-vpc, cohort-subnet-public1-...)
- 4 IPv4 CIDR block: **10.0.0.0/16** (default).
- 5 Number of Availability Zones: **2**. Public subnets: **2**. Private subnets: **0**.
- 6 NAT gateways: **None**. VPC endpoints: **None**.
- 7 DNS options: **Enable DNS hostnames** aur **Enable DNS resolution** dono tick.
- 8 **Create VPC** → complete hone par **View VPC**.

Ye automatically banata hai: 1 VPC, 2 public subnets (2 alag AZ me), 1 Internet Gateway, aur ek route table (`0.0.0.0/0` → **Internet Gateway**).

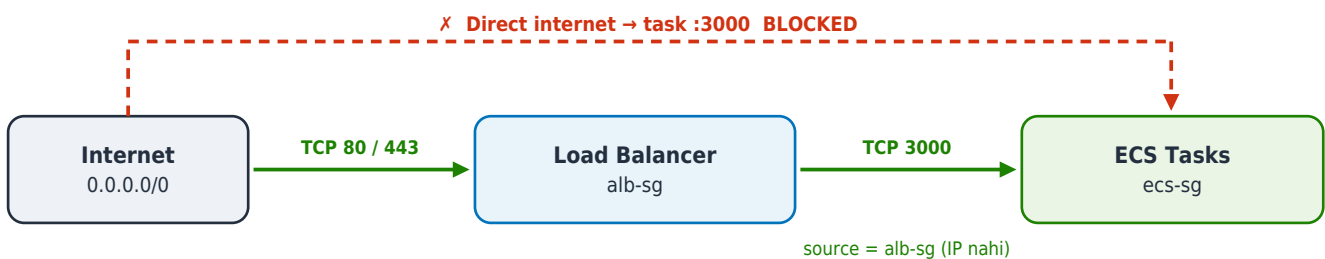
KYUN? (Why)

Private subnet kyun hataya? ALB ko kam se kam **2 subnets (2 alag AZ)** chahiye, wo hamare 2 public subnets de dete hain. Private subnet ke tasks ko ECR / internet tak jaane ke liye **NAT Gateway** chahiye, jo har mahine lagbhag **\$30+** kha jata hai. Learning ke liye tasks public subnet me chalte hain (public IP ke saath) aur **security group** unhe protect karta hai. Production me private subnets + NAT (ya VPC endpoints) use karte hain.

5 Security Groups banao (firewall rules)

Ye pehle isliye banate hain taaki ALB aur Service banate waqt dropdown me mil jayein. Console → **VPC** (ya EC2) → **Security groups** → **Create security group**. **VPC: cohort-vpc** select karna mat bhoolna.

Field	Value / Kya karna hai
SG 1 name / description	alb-sg / "ALB public access"
alb-sg Inbound rule	Type HTTP , port 80 , Source Anywhere-IPv4 (0.0.0.0/0) . (Step 13 me HTTPS 443 bhi add hoga)
SG 2 name / description	ecs-sg / "Only ALB can reach tasks"
ecs-sg Inbound rule	Type Custom TCP , port 3000 , Source = Custom → alb-sg select karo (IP nahi, SG choose karo)
Outbound (dono)	Default (All traffic) rehne do, taaki task ECR / internet tak ja sake



NOTES ME SUDHAAR

Notes me "add security group allow all tcp" likha tha. Wo chalega, par **kamzor** hai: koi bhi internet se seedha task ke public IP par hit kar sakta hai aur ALB ko bypass kar deta hai. Upar wala setup (ecs-sg sirf alb-sg se port 3000 sunta hai) sahi tareeka hai. Debug ke liye temporary rule "My IP" se port 3000 ka laga sakte ho, baad me hata do.

6 ECS Cluster banao

- 1 Console → **ECS** → **Clusters** → **Create cluster**.
- 2 Cluster name: **cohort-cluster**.
- 3 Infrastructure: **AWS Fargate (serverless)** tick rakho (default). Baaki sab default.
- 4 **Create**. (Pehli baar ECS service-linked role apne aap ban sakta hai, normal hai.)

KYUN? (Why)

Cluster sirf ek **logical group** hai: khud me kuch run nahi karta aur khud ka koi charge nahi. Paisa tab lagta hai jab andar tasks chalte hain.

7 Task Execution Role banao

- 1 Console → **IAM** → **Roles** → **Create role**.
- 2 Trusted entity type: **AWS service**. Use case: **Elastic Container Service** → **Elastic Container Service Task** → **Next**.
- 3 Permission search me **AmazonECSTaskExecutionRolePolicy** tick karo → **Next**.
- 4 Role name: **cohort-task-execution-role** → **Create role**.

KYUN? (Why)

Ye role kyun? Container start hone se pehle ECS (Fargate agent) ko tumhari taraf se do kaam karne hote hain: **(1) ECR se image pull** karna aur **(2) CloudWatch me logs likhna**. Uske liye permission yehi role deta hai. Iske bina task **CannotPullContainerError** ke saath fail hota hai.

	Task Execution Role	Task Role
Kaun use karta hai	ECS agent (container start hone se pehle)	Tumhara app code (container ke andar)
Kaam	Image pull, logs likhna, secrets fetch karna	App se S3, DynamoDB, SES jaise AWS services call karna
Abhi?	Zaroori	None rakho (jab tak app AWS services call na kare)

NOTES ME SUDHAAR

Notes me isko "*cohort task role*" likha tha. Naam confuse kar sakta hai, isliye **cohort-task-execution-role** rakho. Shortcut: Task Definition banate waqt ECS console "*Create new role*" se **ecsTaskExecutionRole** khud bana deta hai, wo bhi chalega.

8 Task Definition banao (blueprint)

ECS → **Task definitions** → **Create new task definition** (not JSON; form wala).

Field	Value / Kya karna hai
Task definition family	cohort-task
Launch type	AWS Fargate
OS / Architecture	Linux / X86_64 (Step 3 ke amd64 image se match hona chahiye)
Network mode	awsvpc (Fargate me fixed)
Task size	CPU 0.5 vCPU , Memory 1 GB (demo ke liye kaafi)
Task role	None
Task execution role	cohort-task-execution-role (Step 7 wala)
Container: Name	cohort-demo-server
Container: Image URI	<ACCOUNT_ID>.dkr.ecr.ap-south-1.amazonaws.com/cohort_demo:latest (ya Browse ECR images se choose karo)
Container: Port mappings	Container port 3000 , Protocol TCP , App protocol HTTP
Environment variables	Key-value daalo, e.g. NODE_ENV=production , PORT=3000 . Secrets ke liye ValueFrom (Secrets Manager / SSM Parameter Store)
Logging	Use log collection ON (awslogs) → CloudWatch log group /ecs/cohort-task

Neeche **Create** dabao. Task definition **cohort-task:1** (revision 1) ban jayegi. Har edit par naya revision banta hai.

KYUN? (Why)

32 GB RAM kyun nahi? Notes me "32 gb ram" example tha. Fargate me 32 GB RAM ke liye kam se kam **8 vCPU** chahiye aur bill size ke hisaab se badhta hai. Ek Express API ke liye 0.25–0.5 vCPU aur 0.5–1 GB kaafi hota hai. Zarurat pade to baad me size badha lo.

✓ TIP

Task Definition sirf **blueprint** hai. Isse bana lene se kuch run nahi hota aur koi charge nahi lagta.

⚠ DHYAN DO

Secrets Manager use karo to **execution role** ko **secretsmanager:GetSecretValue** permission bhi dena padega, warna task start nahi hoga.

9**Target Group + Application Load Balancer**

Hum ALB **pehle alag se** banayenge (EC2 console me), phir Service me use karenge. Isse alb-sg exactly wahi lagta hai jo humne Step 5 me banaya.

9A. Target Group

Field	Value / Kya karna hai
Path	Console → EC2 → left menu Target Groups → Create target group
Target type	IP addresses (Fargate / awsvpc ke liye yehi chahiye)
Name	cohort-tg
Protocol : Port	HTTP : 3000 , IP address type IPv4 , VPC cohort-vpc , Protocol version HTTP1
Health check	Protocol HTTP , Path /health
Register targets	Kuch mat add karo , seedha Create target group . Tasks ko ECS khud register karega

9B. Application Load Balancer

Field	Value / Kya karna hai
Path	EC2 → Load Balancers → Create load balancer → Application Load Balancer → Create
Name / Scheme	cohort-alb / Internet-facing , IPv4
Network mapping	VPC cohort-vpc , dono AZ tick karo aur dono public subnets select karo
Security groups	Default hatao, sirf alb-sg select karo
Listener	HTTP : 80 → Default action Forward to cohort-tg

Create load balancer → State **Active** hone tak 2–3 minute wait. Phir ALB kholo aur **DNS name** copy karo (jaise **cohort-alb-123456789.ap-south-1.elb.amazonaws.com**). Ye aage test aur domain ke kaam aayega.

✓ TIP

Abhi ALB khologe to **503 Service Temporarily Unavailable** dikhega: normal hai, kyunki Target Group me abhi koi task registered nahi hai.

NOTES ME SUDHAAR

Notes me ALB service banate waqt "*create target group*" se banaya tha aur baad me "*allow all tcp*" karna pada. ECS ke "new load balancer" route me ALB ke security group par control kam milta hai, isliye hum ALB alag banate hain.

10 ECS Service banao (2 tasks + ALB)

ECS → **Clusters** → **cohort-cluster** → **Services** tab → **Create**.

Field	Value / Kya karna hai
Compute options	Launch type → FARGATE , Platform version LATEST
Application type	Service
Task definition	Family cohort-task , Revision latest
Service name	cohort-service
Service type / Desired tasks	Replica / 2
Networking: VPC	cohort-vpc
Networking: Subnets	Sirf 2 public subnets rakho (extra ho to hata do)
Networking: Security group	Use an existing → sirf ecs-sg (default wala hatao)
Networking: Public IP	Turned ON
Load balancing: type	Application Load Balancer → Use an existing load balancer → cohort-alb
Load balancing: container	cohort-demo-server 3000:3000
Listener / Target group	Use existing listener 80:HTTP / Use existing target group cohort-tg
Health check grace period	60 seconds

Create dabao. Deployment me 3–5 minute lagte hain. Service → **Tasks** tab me status **PROVISIONING** → **PENDING** → **RUNNING** dikhega. Target Group → **Targets** me dono IP pehle *initial* phir **healthy** honge.

KYUN? (Why)

Desired tasks 2 kyun? Ek task crash ho jaye ya ek AZ me problem aaye to doosra traffic sambhal leta hai. Naya version deploy karte waqt bhi zero-downtime milta hai (naye tasks pehle healthy hote hain, phir purane hatte hain). Kharcha bachana ho to 1 kar do.

KYUN? (Why)

Public IP ON kyun? Tasks public subnet me hain aur NAT nahi hai. Public IP ke bina wo ECR se image pull aur CloudWatch par logs bhej hi nahi paate (error: *CannotPullContainerError* / *ResourceInitializationError*). Bahar se unhe koi seedha nahi pahunch sakta kyunki ecs-sg sirf ALB ko allow karta hai.

11 Test karo (ALB DNS se)

- 1 EC2 → **Load Balancers** → **cohort-alb** → **DNS name** copy karo.
- 2 Browser me kholo: **http://<ALB-DNS>** aur **http://<ALB-DNS>/health** (abhi **http**, https nahi).
- 3 App ka response dikhe to **deployment successful**. Ye step domain lene se pehle hi kar lo, taaki DNS ke jhanjhat se pehle app confirm ho jaye.
- 4 Na chale to **Troubleshooting table** dekho (Step 14 ke baad).

12 Domain ko ALB se jodo

Domain khareed lo (Route 53, GoDaddy, Namecheap, Hostinger, etc.). Phir apne case ke hisaab se neeche ka option follow karo.

Option A: Domain Route 53 me hai (recommended)

- 1 Route 53 me domain register karte hi **Hosted zone** apne aap ban jata hai (nahi bana to **Create hosted zone**).
- 2 Route 53 → **Hosted zones** → apna domain → **Create record**.
- 3 Record name: **khali chhodo** (root domain ke liye) ya **api** (subdomain). Record type: **A**.
- 4 **Alias** toggle **ON** → Route traffic to: **Alias to Application and Classic Load Balancer** → Region **Asia Pacific (Mumbai)** → **cohort-alb** select karo.
- 5 **Create records**.

Option B: Domain GoDaddy / Namecheap / Hostinger jaise registrar par hai

- 1 Registrar ke dashboard me **DNS / Manage DNS** kholo → **Add record**.
- 2 Type **CNAME**, Host / Name: **api** (ya **www**), Value / Points to: **ALB DNS name** (bina http:// ke), TTL 300 / Auto.
- 3 Save karo. Root domain (@) par aam taur par CNAME allowed nahi hota: ya to **subdomain** use karo, ya registrar me **nameservers Route 53 ke 4 NS values** se badal do aur phir Option A follow karo.

Check: `nslookup api.yourdomain.com` ya browser me `http://api.yourdomain.com`. DNS propagation me 5 minute se 48 ghante tak lag sakte hain.

KYUN? (Why)

CNAME ya Alias? ALB ke IP badalte rehte hain, sirf uska DNS name stable hai. Isliye A record me IP nahi, **Alias** (Route 53) ya **CNAME** (baaki registrar) use karte hain. Alias root domain par bhi chalta hai aur Route 53 me uski query free hoti hai.

✓ TIP

Route 53 hosted zone ka lagbhag **\$0.50 / mahina** lagta hai (domain registration alag).

13 HTTPS lagao (notes me nahi tha, par zaroori hai)

13A. Free certificate (ACM)

- 1 Console → **Certificate Manager (ACM)**. Region **Mumbai (ap-south-1)** hi rakho, ALB ke region wala hi certificate use hota hai.
- 2 **Request** → **Request a public certificate** → Domain name **api.yourdomain.com** (chaho to ***.yourdomain.com** bhi).
- 3 Validation method: **DNS validation** → **Request**.
- 4 Certificate kholo → Route 53 me ho to **Create records in Route 53** dabao. Doosre registrar me ho to dikhaya gaya **CNAME name / value** wahan DNS me add karo.
- 5 Status **Issued** hone tak wait (kuch minute).

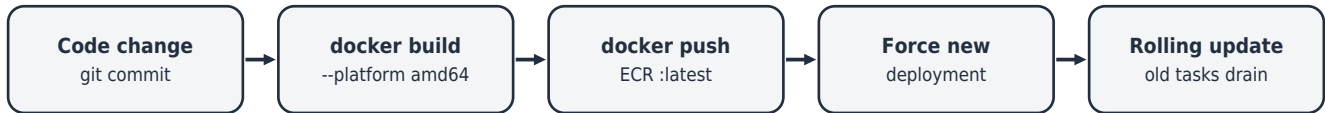
13B. ALB par HTTPS listener

- 1 **alb-sg** me inbound rule add karo: **HTTPS, port 443, 0.0.0.0/0**.
- 2 EC2 → Load Balancers → **cohort-alb** → **Listeners and rules** → **Add listener**.
- 3 Protocol **HTTPS : 443** → Forward to **cohort-tg** → Certificate: **ACM** se apna certificate select → **Add**.
- 4 (Optional, recommended) **HTTP:80** listener edit karo: Default action **Redirect to HTTPS 443** (301).

✓ **TIP**

ALB par HTTPS khatam ho jata hai (TLS termination); ALB se task tak traffic private network me HTTP port 3000 par jata hai. Isliye container me kuch badalna nahi padta. ACM certificate free hai aur DNS validation record rakhne par auto-renew hota hai.

14 Naya code deploy karo (redeploy)



- 1 Code badlo → Step 3 ke **build, tag, push** commands dobara chalao (login ki zarurat sirf token expire hone par).
- 2 ECS → cluster → **cohort-service** → **Update service** → **Force new deployment** tick → **Update**.
- 3 Rolling update: naye tasks start hote hain, health check pass karte hain, phir purane tasks drain hokar band ho jaate hain.

```
aws ecs update-service --cluster cohort-cluster --service cohort-service \
  --force-new-deployment --region ap-south-1
```

KYUN? (Why)

Force new deployment kyun? Tag `:latest` wahi rehta hai, isliye ECS ko pata hi nahi chalta ki image badli hai. Force karne par wo naye tasks par image dobara pull karta hai. Better practice: har push par unique tag (jaise git commit id `:a1b2c3d`) do aur Task Definition ka naya revision banao.

✓ **TIP**

Logs dekhne ke liye: ECS → service → **Logs** tab, ya CloudWatch → Log groups → `/ecs/cohort-task`.

Troubleshooting (jab kuch na chale)

Symptom	Wajah	Fix
docker push : no basic auth credentials	ECR login nahi hua / token expire	Step 3 ka command 1 dobara chalo
AccessDenied / not authorized	IAM user ke paas policy nahi	Step 1 ki table se policy attach karo (ECR policy mat bhoolna)
Task turant stop, logs me <i>exec format error</i>	Image arm64 bani (Apple Silicon)	Rebuild: --platform linux/amd64 , phir push + force deploy
<i>CannotPullContainerError / ResourceInitializationError</i>	Public IP OFF, ya execution role nahi, ya image URI / tag galat	Service me Public IP ON, role attach, URI check
Task <i>essential container exited</i>	App crash (env var missing, DB connect fail, galat start command)	CloudWatch logs padho
Targets unhealthy	Health path 200 nahi de raha, port galat, ecs-sg me alb-sg se 3000 allowed nahi	Path /health , port 3000, ecs-sg inbound rule check
ALB par 503	Target group me koi healthy target nahi	Service RUNNING? Targets healthy? Wait karo
ALB par 502	App ne connection band kiya / sirf localhost par sunti hai	App ko sab interfaces par listen karao
ALB par 504 / timeout	Security group / subnet route dikkat, ya app bahut slow	alb-sg 80/443 open, route table me IGW, app response time
Domain nahi khul raha	DNS propagation, galat record type, root par CNAME, cert se pehle https	nslookup karo, Alias / CNAME sahi karo, wait karo
Cert Pending validation	DNS validation CNAME add nahi hua	ACM me dikhaaya CNAME DNS me daalo

Cleanup: bill se bachne ke liye (kaam khatam hone par)

- 1 ECS → service → **Update**: desired tasks **0** → phir service **Delete**.
- 2 ECS → **cohort-cluster** → Delete cluster.
- 3 EC2 → **Load Balancers** → cohort-alb → Delete. Phir **Target Groups** → cohort-tg → Delete.
- 4 VPC → **cohort-vpc** → Delete VPC (subnets, IGW, route table saath me hat jaate hain). Pehle alb-sg aur ecs-sg delete karne pad sakte hain.
- 5 ECR → **cohort_demo** → images / repository delete. CloudWatch → log group **/ecs/cohort-task** delete.
- 6 Route 53 me record / hosted zone aur ACM certificate hata do (agar ab nahi chahiye).
- 7 IAM → cohort_user → **access key Delete / Deactivate** (jab tak kaam nahi hai).

⚠ DHYAN DO

Sabse zyada idle charge: ALB (hourly), Fargate tasks (vCPU + GB per hour) aur **public IPv4 addresses** (har task + ALB ka, lagbhag \$0.005 / ghanta each). Sirf tasks 0 karne se ALB ka bill chalta rehta hai, isliye ALB bhi delete karo.

Naam cheat sheet (is guide me jo naam use hue)

Cheez	Naam / Value
Region	ap-south-1 (Mumbai)
IAM user	cohort_user
ECR repository	cohort_demo → <ACCOUNT_ID>.dkr.ecr.ap-south-1.amazonaws.com/cohort_demo:latest
VPC	cohort-vpc (10.0.0.0/16, 2 public subnets, 0 private, no NAT)

Cheez	Naam / Value
Security groups	alb-sg (80/443 from anywhere), ecs-sg (3000 from alb-sg)
ECS cluster	cohort-cluster (Fargate)
Execution role	cohort-task-execution-role (AmazonECSTaskExecutionRolePolicy)
Task definition	cohort-task container cohort-demo-server port 3000 0.5 vCPU / 1 GB Linux X86_64
Target group / ALB	cohort-tg (IP, HTTP:3000, /health) / cohort-alb (internet-facing, public subnets, alb-sg)
ECS service	cohort-service desired 2 public IP ON ecs-sg
Domain	api.yourdomain.com → ALB (Alias A in Route 53, ya CNAME)

Tumhare sawaalon ke jawab (quick FAQ)

Sawaal	Short jawab	Step
Root user kyun nahi?	Root ka access unlimited hai; leak = poora account. IAM user limited hota hai aur key delete ho sakti hai	1
Warp terminal kyun?	Zaroori nahi, sirf ek achha terminal hai. Koi bhi terminal chalega	2
Windows par Mac wale commands?	Windows tab PowerShell ke liye hai; Git Bash me Mac / Linux commands chalte hain	3
2nd command kyun nahi chalana?	Normal build arm64 image bana sakta hai; Fargate x86 hai. buildx + --platform linux/amd64 use karo	3
Private subnet kyun remove?	NAT Gateway mehenga hai; learning me public subnet + security group kaafi	4
"Allow all TCP" theek hai?	Chalega par unsafe. ecs-sg me sirf alb-sg se port 3000 allow karo	5
Task execution role kya hai?	ECS ko ECR se image pull aur logs likhne ki permission deta hai	7
Task definition vs service?	Definition = blueprint (recipe). Service = usse tasks chalata aur zinda rakhta hai	8, 10
Desired tasks 2 kyun?	High availability + zero-downtime deploy	10
Target group kya hai?	ALB ke peeche ke tasks ki list + health check	9
CNAME ya Alias?	Route 53 me Alias, baaki registrar me CNAME (root par nahi)	12
Pehle http kyun, https nahi?	Certificate Step 13 me aata hai; pehle http se confirm karo app chal rahi hai	11, 13